

TEAM ID	SWTID-2026-8827
PROJECT TITLE	India's Agricultural Crop Production Analysis.
MAX MARKS	3 MARKS

3.4 – Technology Stack

The technology stack for India's Agricultural Crop Production Analysis system includes various tools and technologies used to collect, store, process, analyse, and visualize agricultural data. These technologies help improve crop productivity, support decision-making, and promote sustainable farming practices.

1. Data Collection Layer

- IoT Sensors
- Weather Stations
- Satellite & Remote Sensing Data
- GPS Devices
- Mobile Applications
- Government Agricultural Databases

2. Data Storage Layer

- MySQL
- PostgreSQL
- MongoDB
- Hadoop Distributed File System (HDFS)
- Cloud Storage Services

3. Data Processing Layer

- Apache Hadoop
- Apache Spark
- Python
- R Programming
- ETL Tools

4. Analytics & Machine Learning Layer

- Scikit-learn
- TensorFlow
- Keras
- PyTorch
- Crop Yield Prediction Models
- Disease Detection Models

5. GIS Layer

- QGIS
- ArcGIS
- Google Earth Engine
- Remote Sensing Technologies

6. Visualization & Reporting Layer

- Power BI
- Tableau
- Matplotlib
- Seaborn
- Plotly

7. Application Development Layer

Frontend: HTML5, CSS3, JavaScript, Bootstrap, React.js

Backend: Python, Node.js, REST APIs

8. Cloud & Deployment Layer

- AWS
- Microsoft Azure
- Google Cloud Platform
- Docker
- Kubernetes

9. Security Layer

- Data Encryption
- User Authentication & Authorization
- Secure API Access
- Cloud Security Services

Benefits

- Improves crop yield prediction and forecasting.
- Supports weather-based farming decisions.